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Roll No: 24

MCA_III

Mobile Application Development

Practical Assignment - I

1. **Create an app with splash activity as your launcher activity, put some view or frame animation into it.**

```
<!-- File: AndroidManifest.xml -->
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

   <application
      android:allowBackup="true"
      android:theme="@style/Theme.DemoApplication"
      android:label="Splash App"
      android:supportsRtl="true">

      <activity
         android:name=".MainActivity"
         android:exported="false" />

      <activity
         android:name=".SplashActivity"
         android:exported="true">

         <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
         </intent-filter>

      </activity>
```

```
    </application>

</manifest>



// File: SplashActivity.java

package com.example.demoapplication;

import android.app.Activity;
import android.content.Intent;
import android.os.Bundle;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.ImageView;
import android.widget.TextView;

public class SplashActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_splash);

        ImageView logo = findViewById(R.id.logo);
        TextView title = findViewById(R.id.title);

        Animation animation = AnimationUtils.loadAnimation(this,
R.anim.splash_animation);

        logo.startAnimation(animation);
        title.startAnimation(animation);

        new android.os.Handler().postDelayed(() -> {
            startActivity(new Intent(SplashActivity.this, MainActivity.class));
            finish();
        }, 3000);
    }
```

```
}



// File: MainActivity.java

package com.example.demoapplication;

import android.app.Activity;
import android.os.Bundle;
import android.widget.TextView;
import android.view.Gravity;

public class MainActivity extends Activity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        TextView textView = new TextView(this);
        textView.setText("Welcome to My Application!");
        textView.setTextSize(24);
        textView.setGravity(Gravity.CENTER);

        setContentView(textView);
    }
}



<!-- File: res/layout/activity_splash.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:background="#2196F3">

    <ImageView
```

```
        android:id="@+id/logo"
        android:layout_width="120dp"
        android:layout_height="120dp"
        android:src="@android:drawable/ic_dialog_info"
        android:contentDescription="App Logo"
        android:tint="#FFFFFF" />

    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Application"
        android:textColor="#FFFFFF"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp" />

</LinearLayout>


<!-- File: res/anim/splash_animation.xml -->

<?xml version="1.0" encoding="utf-8"?>
<set xmlns:android="http://schemas.android.com/apk/res/android"
    android:interpolator="@android:anim/accelerate_decelerate_interpolator">

    <alpha
        android:fromAlpha="0.0"
        android:toAlpha="1.0"
        android:duration="2000" />

    <scale
        android:fromXScale="0.5"
        android:toXScale="1.0"
        android:fromYScale="0.5"
        android:toYScale="1.0"
        android:pivotX="50%"
        android:pivotY="50%"
        android:duration="2000" />
```

</set>

1. Design user role based login and home screen app
    - Normal User Login
        - Login with : EmailId as user@user.com and Password as user@12345
        - Home screen (Form to enter employee data and Logout button)
            - Name
            - EmailId
            - Age
            - Gender
            - Salary
        - verify employee credentials from Shared Preferences

```
// File: MainActivity.java

package com.example.demoapplication;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   EditText email, password;
```

```
Button login;
SharedPreferences preferences;

@Override
protected void onCreate(Bundle savedInstanceState) {
   super.onCreate(savedInstanceState);

   preferences = getSharedPreferences("UserPrefs", MODE_PRIVATE);
   preferences.edit()
         .putString("email", "user@user.com")
         .putString("password", "user@12345")
         .apply();

   if (preferences.getBoolean("loggedIn", false)) {
      startActivity(new Intent(this, HomeActivity.class));
      finish();
      return;
   }

   setContentView(R.layout.activity_main);

   email = findViewById(R.id.email);
   password = findViewById(R.id.password);
   login = findViewById(R.id.login);

   login.setOnClickListener(v -> {
      String enteredEmail = email.getText().toString().trim();
      String enteredPassword = password.getText().toString();

      String savedEmail = preferences.getString("email", "");
      String savedPassword = preferences.getString("password", "");

      if (enteredEmail.equals(savedEmail)
            && enteredPassword.equals(savedPassword)) {

         preferences.edit().putBoolean("loggedIn", true).apply();

         startActivity(new Intent(this, HomeActivity.class));
         finish();
      } else {
```

```
                Toast.makeText(this, "Invalid Email or Password",
                        Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

```
<!-- File: res/layout/activity_main.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Normal User Login"
        android:textSize="26sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp" />

    <EditText
        android:id="@+id/email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email ID"
        android:inputType="textEmailAddress" />

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword" />
```

```
    <Button
        android:id="@+id/login"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="16dp" />

</LinearLayout>
```

```
// File: HomeActivity.java

package com.example.demoapplication;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioGroup;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class HomeActivity extends AppCompatActivity {

    EditText name, email, age, salary;
    RadioGroup gender;
    Button save, logout;
    SharedPreferences preferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        preferences = getSharedPreferences("UserPrefs", MODE_PRIVATE);

        if (!preferences.getBoolean("loggedIn", false)) {
```

```
    startActivity(new Intent(this, MainActivity.class));
    finish();
    return;
}

setContentView(R.layout.activity_home);

name = findViewById(R.id.name);
email = findViewById(R.id.employeeEmail);
age = findViewById(R.id.age);
salary = findViewById(R.id.salary);
gender = findViewById(R.id.gender);
save = findViewById(R.id.save);
logout = findViewById(R.id.logout);

save.setOnClickListener(v -> {
    String employeeName = name.getText().toString().trim();
    String employeeEmail = email.getText().toString().trim();
    String employeeAge = age.getText().toString().trim();
    String employeeSalary = salary.getText().toString().trim();

    if (employeeName.isEmpty() || employeeEmail.isEmpty()
            || employeeAge.isEmpty() || employeeSalary.isEmpty()
            || gender.getCheckedRadioButtonId() == -1) {
        Toast.makeText(this, "Please fill all fields",
                Toast.LENGTH_SHORT).show();
        return;
    }

    String selectedGender = gender.getCheckedRadioButtonId() == R.id.male
            ? "Male" : "Female";

    preferences.edit()
            .putString("employeeName", employeeName)
            .putString("employeeEmail", employeeEmail)
            .putString("employeeAge", employeeAge)
            .putString("employeeGender", selectedGender)
            .putString("employeeSalary", employeeSalary)
            .apply();
```

```
            Toast.makeText(this, "Employee data saved",
                    Toast.LENGTH_SHORT).show();
        });

        logout.setOnClickListener(v -> {
            preferences.edit().putBoolean("loggedIn", false).apply();
            startActivity(new Intent(this, MainActivity.class));
            finish();
        });
    }
}
```

```
<!-- File: res/layout/activity_home.xml -->

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Employee Home Screen"
            android:textSize="24sp"
            android:textStyle="bold"
            android:layout_gravity="center"
            android:layout_marginBottom="20dp" />

        <EditText
            android:id="@+id/name"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
    android:hint="Employee Name"
    android:inputType="textPersonName" />

<EditText
    android:id="@+id/employeeEmail"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Employee Email ID"
    android:inputType="textEmailAddress" />

<EditText
    android:id="@+id/age"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Age"
    android:inputType="number" />

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Gender"
    android:textSize="16sp"
    android:layout_marginTop="12dp" />

<RadioGroup
    android:id="@+id/gender"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <RadioButton
        android:id="@+id/male"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male" />

    <RadioButton
        android:id="@+id/female"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
            android:text="Female" />

        </RadioGroup>

        <EditText
            android:id="@+id/salary"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Salary"
            android:inputType="numberDecimal" />

        <Button
            android:id="@+id/save"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Save Employee Data"
            android:layout_marginTop="16dp" />

        <Button
            android:id="@+id/logout"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Logout" />

    </LinearLayout>
</ScrollView>



<!-- File: AndroidManifest.xml -->

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="Employee App"
        android:supportsRtl="true"
        android:theme="@style/Theme.DemoApplication">
```

```
        <activity
            android:name=".HomeActivity"
            android:exported="false" />

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>
</manifest>
```

3. Create a home screen with three tabs

a. **Tab 1 to display product list of your app**
b. **Tab2 to display gallery images in recyclerview in grid manager**
c. **Tab 3 to display user profile detail where he/she can update his/her details**

```
// File: MainActivity.java


package com.example.demoapplication;


import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.viewpager2.widget.ViewPager2;
import com.google.android.material.tabs.TabLayout;
import com.google.android.material.tabs.TabLayoutMediator;

public class MainActivity extends AppCompatActivity {

    TabLayout tabLayout;
    ViewPager2 viewPager;
    ViewPagerAdapter adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tabLayout = findViewById(R.id.tabLayout);
        viewPager = findViewById(R.id.viewPager);

        adapter = new ViewPagerAdapter(this);
        viewPager.setAdapter(adapter);

        new TabLayoutMediator(tabLayout, viewPager, (tab, position) -> {
            switch (position) {
                case 0:
```

```
                tab.setText("Products");
                break;
            case 1:
                tab.setText("Gallery");
                break;
            case 2:
                tab.setText("Profile");
                break;
        }
    }).attach();
  }
}
```

```
<!-- File: res/layout/activity_main.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical">
```

```
    <com.google.android.material.tabs.TabLayout
        android:id="@+id/tabLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        app:tabMode="fixed"
        app:tabGravity="fill" />

    <androidx.viewpager2.widget.ViewPager2
        android:id="@+id/viewPager"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

```
// File: ViewPagerAdapter.java

package com.example.demoapplication;

import androidx.annotation.NonNull;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentActivity;
import androidx.viewpager2.adapter.FragmentStateAdapter;
```

```
public class ViewPagerAdapter extends FragmentStateAdapter {

    public ViewPagerAdapter(@NonNull FragmentActivity activity) {
        super(activity);
    }

    @NonNull
    @Override
    public Fragment createFragment(int position) {
        switch (position) {
            case 0:
                return new ProductFragment();
            case 1:
                return new GalleryFragment();
            default:
                return new ProfileFragment();
        }
    }

    @Override
    public int getItemCount() {
        return 3;
    }
```

```
}



// File: ProductFragment.java

package com.example.demoapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class ProductFragment extends Fragment {

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
```

```
                @Nullable Bundle savedInstanceState) {

View view = inflater.inflate(R.layout.fragment_product, container, false);

ListView listView = view.findViewById(R.id.productList);

String[] products = {
        "Laptop - ₹50,000",
        "Mobile Phone - ₹20,000",
        "Headphones - ₹2,000",
        "Smart Watch - ₹3,000",
        "Keyboard - ₹1,500",
        "Mouse - ₹800"
};

ArrayAdapter<String> adapter = new ArrayAdapter<>(
        requireContext(),
        android.R.layout.simple_list_item_1,
        products
);

listView.setAdapter(adapter);

return view;
```

```
    }
}
```

```
<!-- File: res/layout/fragment_product.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Product List"
        android:textSize="24sp"
        android:textStyle="bold"
        android:padding="8dp" />

    <ListView
        android:id="@+id/productList"
```

```
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>



// File: GalleryFragment.java

package com.example.demoapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
import androidx.recyclerview.widget.GridLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

public class GalleryFragment extends Fragment {

    @Nullable
```

```
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
                             @Nullable Bundle savedInstanceState) {

        View view = inflater.inflate(R.layout.fragment_gallery, container, false);

        RecyclerView recyclerView = view.findViewById(R.id.galleryRecyclerView);
        recyclerView.setLayoutManager(new GridLayoutManager(requireContext(), 2));

        int[] images = {
                R.drawable.gallery1,
                R.drawable.gallery2,
                R.drawable.gallery3,
                R.drawable.gallery4,
                R.drawable.gallery5,
                R.drawable.gallery6
        };

        recyclerView.setAdapter(new GalleryAdapter(images));

        return view;
    }
}
```

```
// File: GalleryAdapter.java

package com.example.demoapplication;

import android.view.View;
import android.view.ViewGroup;
import android.view.LayoutInflater;
import android.widget.ImageView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class GalleryAdapter extends RecyclerView.Adapter<GalleryAdapter.ImageViewHolder> {

    private final int[] images;

    public GalleryAdapter(int[] images) {
        this.images = images;
    }

    @NonNull
```

```
    @Override
    public ImageViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
        ImageView imageView = new ImageView(parent.getContext());

        imageView.setLayoutParams(new ViewGroup.LayoutParams(
                ViewGroup.LayoutParams.MATCH_PARENT, 250));

        imageView.setScaleType(ImageView.ScaleType.CENTER_CROP);
        imageView.setPadding(6, 6, 6, 6);

        return new ImageViewHolder(imageView);
    }

    @Override
    public void onBindViewHolder(@NonNull ImageViewHolder holder, int position)
{
        holder.imageView.setImageResource(images[position]);
    }

    @Override
    public int getItemCount() {
        return images.length;
    }
```

```
    static class ImageViewHolder extends RecyclerView.ViewHolder {

        ImageView imageView;

        ImageViewHolder(@NonNull View itemView) {
            super(itemView);
            imageView = (ImageView) itemView;
        }
    }
}
```

```
<!-- File: res/layout/fragment_gallery.xml -->

<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="8dp">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/galleryRecyclerView"
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent" />

</FrameLayout>



// File: ProfileFragment.java

package com.example.demoapplication;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class ProfileFragment extends Fragment {
```

```
EditText name, email, phone;
SharedPreferences preferences;

@Nullable
@Override
public View onCreateView(@NonNull LayoutInflater inflater,
                         @Nullable ViewGroup container,
                         @Nullable Bundle savedInstanceState) {

    View view = inflater.inflate(R.layout.fragment_profile, container, false);

    name = view.findViewById(R.id.profileName);
    email = view.findViewById(R.id.profileEmail);
    phone = view.findViewById(R.id.profilePhone);
    Button updateButton = view.findViewById(R.id.updateProfile);

    preferences = requireActivity().getSharedPreferences(
            "ProfilePrefs", android.content.Context.MODE_PRIVATE);

    name.setText(preferences.getString("name", ""));
    email.setText(preferences.getString("email", ""));
    phone.setText(preferences.getString("phone", ""));

    updateButton.setOnClickListener(v -> {
```

```
            String userName = name.getText().toString().trim();
            String userEmail = email.getText().toString().trim();
            String userPhone = phone.getText().toString().trim();

            if (userName.isEmpty() || userEmail.isEmpty() || userPhone.isEmpty()) {
                Toast.makeText(requireContext(), "Please fill all fields",
                        Toast.LENGTH_SHORT).show();
                return;
            }

            preferences.edit()
                    .putString("name", userName)
                    .putString("email", userEmail)
                    .putString("phone", userPhone)
                    .apply();

            Toast.makeText(requireContext(), "Profile updated successfully",
                    Toast.LENGTH_SHORT).show();
        });

        return view;
    }
}
```

```
<!-- File: res/layout/fragment_profile.xml -->

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="User Profile"
            android:textSize="24sp"
            android:textStyle="bold"
            android:paddingBottom="20dp" />

        <EditText
```

```
    android:id="@+id/profileName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Full Name"
    android:inputType="textPersonName" />

<EditText
    android:id="@+id/profileEmail"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Email Address"
    android:inputType="textEmailAddress" />

<EditText
    android:id="@+id/profilePhone"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Phone Number"
    android:inputType="phone" />

<Button
    android:id="@+id/updateProfile"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:text="Update Profile"
        android:layout_marginTop="20dp" />

    </LinearLayout>
</ScrollView>



<!-- File: AndroidManifest.xml -->

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="Tabbed App"
        android:supportsRtl="true"
        android:theme="@style/Theme.DemoApplication">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
```

```
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>

    </activity>

</application>
</manifest>



// File: app/build.gradle

dependencies {
    implementation 'androidx.appcompat:appcompat:1.7.0'
    implementation 'androidx.fragment:fragment:1.8.5'
    implementation 'androidx.recyclerview:recyclerview:1.3.2'
    implementation 'androidx.viewpager2:viewpager2:1.1.0'
    implementation 'com.google.android.material:material:1.12.0'
}



// Required image files:
// File: app/src/main/res/drawable/gallery1.jpg
```

// File: app/src/main/res/drawable/gallery2.jpg

// File: app/src/main/res/drawable/gallery3.jpg

// File: app/src/main/res/drawable/gallery4.jpg

// File: app/src/main/res/drawable/gallery5.jpg

// File: app/src/main/res/drawable/gallery6.jpg

4. Add navigation drawer menu

1. Add product / update product detail

2. Remove Product

2. Contact list

3. Log Out

// File: MainActivity.java

```
package com.example.demoapplication;

import android.content.Intent;
import android.os.Bundle;
import android.view.MenuItem;
import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
```

```
import androidx.appcompat.widget.Toolbar;
import androidx.drawerlayout.widget.DrawerLayout;
import com.google.android.material.navigation.NavigationView;

public class MainActivity extends AppCompatActivity
        implements NavigationView.OnNavigationItemSelectedListener {

    DrawerLayout drawerLayout;
    NavigationView navigationView;
    Toolbar toolbar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        drawerLayout = findViewById(R.id.drawerLayout);
        navigationView = findViewById(R.id.navigationView);
        toolbar = findViewById(R.id.toolbar);

        setSupportActionBar(toolbar);

        ActionBarDrawerToggle toggle = new ActionBarDrawerToggle(
                this, drawerLayout, toolbar,
```

```
            R.string.navigation_drawer_open,
            R.string.navigation_drawer_close);

    drawerLayout.addDrawerListener(toggle);
    toggle.syncState();

    navigationView.setNavigationItemSelectedListener(this);

    if (savedInstanceState == null) {
        getSupportFragmentManager().beginTransaction()
                .replace(R.id.contentFrame, new ProductFragment())
                .commit();
        navigationView.setCheckedItem(R.id.nav_product);
    }
}

@Override
public boolean onNavigationItemSelected(@NonNull MenuItem item) {
    int id = item.getItemId();

    if (id == R.id.nav_product) {
        getSupportFragmentManager().beginTransaction()
                .replace(R.id.contentFrame, new ProductFragment())
                .commit();
```

```
        toolbar.setTitle("Add / Update Product");
    } else if (id == R.id.nav_remove) {
        getSupportFragmentManager().beginTransaction()
                .replace(R.id.contentFrame, new RemoveProductFragment())
                .commit();
        toolbar.setTitle("Remove Product");
    } else if (id == R.id.nav_contacts) {
        getSupportFragmentManager().beginTransaction()
                .replace(R.id.contentFrame, new ContactFragment())
                .commit();
        toolbar.setTitle("Contact List");
    } else if (id == R.id.nav_logout) {
        getSharedPreferences("UserPrefs", MODE_PRIVATE)
                .edit().putBoolean("loggedIn", false).apply();

        Intent intent = new Intent(this, MainActivity.class);
        intent.addFlags(Intent.FLAG_ACTIVITY_CLEAR_TOP
                | Intent.FLAG_ACTIVITY_NEW_TASK);
        finish();
    }

    drawerLayout.closeDrawers();
    return true;
}
```

```
}
```

```
<!-- File: res/layout/activity_main.xml -->

<?xml version="1.0" encoding="utf-8"?>
<androidx.drawerlayout.widget.DrawerLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/drawerLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical">

        <androidx.appcompat.widget.Toolbar
            android:id="@+id/toolbar"
            android:layout_width="match_parent"
            android:layout_height="?attr/actionBarSize"
```

```
            android:background="?attr/colorPrimary"
            app:titleTextColor="@android:color/white" />

        <FrameLayout
            android:id="@+id/contentFrame"
            android:layout_width="match_parent"
            android:layout_height="match_parent" />

    </LinearLayout>

    <com.google.android.material.navigation.NavigationView
        android:id="@+id/navigationView"
        android:layout_width="wrap_content"
        android:layout_height="match_parent"
        android:layout_gravity="start"
        app:menu="@menu/drawer_menu" />

</androidx.drawerlayout.widget.DrawerLayout>



<!-- File: res/menu/drawer_menu.xml -->

<?xml version="1.0" encoding="utf-8"?>
```

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/nav_product"
        android:title="Add / Update Product"
        android:icon="@android:drawable/ic_input_add" />

    <item
        android:id="@+id/nav_remove"
        android:title="Remove Product"
        android:icon="@android:drawable/ic_menu_delete" />

    <item
        android:id="@+id/nav_contacts"
        android:title="Contact List"
        android:icon="@android:drawable/ic_menu_call" />

    <item
        android:id="@+id/nav_logout"
        android:title="Log Out"
        android:icon="@android:drawable/ic_lock_power_off" />

</menu>
```

```
// File: ProductFragment.java

package com.example.demoapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class ProductFragment extends Fragment {

    EditText productId, productName, productPrice;
    Button saveButton;

    @Nullable
    @Override
```

```
public View onCreateView(@NonNull LayoutInflater inflater,
                         @Nullable ViewGroup container,
                         @Nullable Bundle savedInstanceState) {

    View view = inflater.inflate(R.layout.fragment_product, container, false);

    productId = view.findViewById(R.id.productId);
    productName = view.findViewById(R.id.productName);
    productPrice = view.findViewById(R.id.productPrice);
    saveButton = view.findViewById(R.id.saveProduct);

    saveButton.setOnClickListener(v -> {
        String id = productId.getText().toString().trim();
        String name = productName.getText().toString().trim();
        String price = productPrice.getText().toString().trim();

        if (id.isEmpty() || name.isEmpty() || price.isEmpty()) {
            Toast.makeText(requireContext(), "Enter all details",
                    Toast.LENGTH_SHORT).show();
            return;
        }

        try {
            double amount = Double.parseDouble(price);
```

```
            if (amount < 0) {
                Toast.makeText(requireContext(), "Invalid price",
                        Toast.LENGTH_SHORT).show();
                return;
            }
        } catch (NumberFormatException e) {
            Toast.makeText(requireContext(), "Enter a valid price",
                    Toast.LENGTH_SHORT).show();
            return;
        }

        ProductStorage.saveProduct(requireContext(), id, name, price);

        Toast.makeText(requireContext(), "Product added/updated",
                Toast.LENGTH_SHORT).show();

        productId.setText("");
        productName.setText("");
        productPrice.setText("");
    });

    return view;
  }
}
```

```
<!-- File: res/layout/fragment_product.xml -->

<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="24dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Add / Update Product"
            android:textSize="24sp"
            android:textStyle="bold"
            android:paddingBottom="20dp" />

        <EditText
```

```
    android:id="@+id/productId"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Product ID"
    android:inputType="text" />

<EditText
    android:id="@+id/productName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Product Name"
    android:inputType="textCapWords" />

<EditText
    android:id="@+id/productPrice"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Product Price"
    android:inputType="numberDecimal" />

<Button
    android:id="@+id/saveProduct"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:text="Save Product"
        android:layout_marginTop="16dp" />

    </LinearLayout>
</ScrollView>
```

```
// File: ProductStorage.java

package com.example.demoapplication;

import android.content.Context;
import android.content.SharedPreferences;
import org.json.JSONArray;
import org.json.JSONException;
import org.json.JSONObject;

public class ProductStorage {

    private static final String PREFS = "ProductPrefs";
    private static final String KEY = "products";

    public static JSONArray getProducts(Context context) {
```

```
    SharedPreferences prefs =
            context.getSharedPreferences(PREFS, Context.MODE_PRIVATE);

    String data = prefs.getString(KEY, "[]");

    try {
        return new JSONArray(data);
    } catch (JSONException e) {
        return new JSONArray();
    }
}

public static void saveProduct(Context context, String id,
                               String name, String price) {
    JSONArray oldProducts = getProducts(context);
    JSONArray updatedProducts = new JSONArray();
    JSONObject product = new JSONObject();

    try {
        product.put("id", id);
        product.put("name", name);
        product.put("price", price);

        boolean updated = false;
```

```
        for (int i = 0; i < oldProducts.length(); i++) {
            JSONObject existing = oldProducts.getJSONObject(i);

            if (existing.getString("id").equals(id)) {
                updatedProducts.put(product);
                updated = true;
            } else {
                updatedProducts.put(existing);
            }
        }

        if (!updated) {
            updatedProducts.put(product);
        }

        context.getSharedPreferences(PREFS, Context.MODE_PRIVATE)
                .edit().putString(KEY, updatedProducts.toString()).apply();

    } catch (JSONException e) {
        e.printStackTrace();
    }
}
```

```
public static boolean removeProduct(Context context, String id) {
    JSONArray oldProducts = getProducts(context);
    JSONArray updatedProducts = new JSONArray();
    boolean removed = false;

    try {
        for (int i = 0; i < oldProducts.length(); i++) {
            JSONObject product = oldProducts.getJSONObject(i);

            if (product.getString("id").equals(id)) {
                removed = true;
            } else {
                updatedProducts.put(product);
            }
        }

        context.getSharedPreferences(PREFS, Context.MODE_PRIVATE)
                .edit().putString(KEY, updatedProducts.toString()).apply();

    } catch (JSONException e) {
        e.printStackTrace();
    }

    return removed;
```

```
    }
}



// File: RemoveProductFragment.java

package com.example.demoapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class RemoveProductFragment extends Fragment {

    @Nullable
    @Override
```

```
public View onCreateView(@NonNull LayoutInflater inflater,
                         @Nullable ViewGroup container,
                         @Nullable Bundle savedInstanceState) {

    View view = inflater.inflate(R.layout.fragment_remove_product,
            container, false);

    EditText productId = view.findViewById(R.id.removeProductId);
    Button removeButton = view.findViewById(R.id.removeProductButton);

    removeButton.setOnClickListener(v -> {
        String id = productId.getText().toString().trim();

        if (id.isEmpty()) {
            productId.setError("Enter Product ID");
            return;
        }

        boolean removed = ProductStorage.removeProduct(requireContext(), id);

        Toast.makeText(requireContext(),
                removed ? "Product removed" : "Product not found",
                Toast.LENGTH_SHORT).show();
```

```
            productId.setText("");
        });

        return view;
    }
}
```

```
<!-- File: res/layout/fragment_remove_product.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Remove Product"
        android:textSize="24sp"
        android:textStyle="bold"
```

```
        android:paddingBottom="20dp" />

    <EditText
        android:id="@+id/removeProductId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Product ID"
        android:inputType="text" />

    <Button
        android:id="@+id/removeProductButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Remove Product"
        android:layout_marginTop="16dp" />

</LinearLayout>



// File: ContactFragment.java

package com.example.demoapplication;
```

```
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class ContactFragment extends Fragment {

    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater,
                             @Nullable ViewGroup container,
                             @Nullable Bundle savedInstanceState) {

        View view = inflater.inflate(R.layout.fragment_contact, container, false);

        ListView contactList = view.findViewById(R.id.contactList);

        String[] contacts = {
                "Dhruv - 9876543210",
```

```
            "Raj - 9876501234",
            "Nirav - 9876512345",
            "Amit - 9876523456",
            "Rahul - 9876534567"
        };

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            requireContext(),
            android.R.layout.simple_list_item_1,
            contacts
        );

        contactList.setAdapter(adapter);

        return view;
    }
}
```

```
<!-- File: res/layout/fragment_contact.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Contact List"
        android:textSize="24sp"
        android:textStyle="bold"
        android:padding="8dp" />

    <ListView
        android:id="@+id/contactList"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>



<!-- File: res/values/strings.xml -->
```

```
<resources>
    <string name="app_name">Navigation Drawer App</string>
    <string name="navigation_drawer_open">Open navigation drawer</string>
    <string name="navigation_drawer_close">Close navigation drawer</string>
</resources>
```

```
// File: app/build.gradle

dependencies {
    implementation 'androidx.appcompat:appcompat:1.7.0'
    implementation 'androidx.fragment:fragment:1.8.5'
    implementation 'androidx.drawerlayout:drawerlayout:1.2.0'
    implementation 'com.google.android.material:material:1.12.0'
}
```

6. Create ToDo list app using recycler view store all the task in table “Task”

Apply insert, delete and view operations using SQLite Database

```
// File: MainActivity.java

package com.example.demoapplication;
```

```
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    EditText taskInput;
    Button addButton;
    RecyclerView recyclerView;
    DatabaseHelper databaseHelper;
    TaskAdapter taskAdapter;
    ArrayList<Task> taskList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        taskInput = findViewById(R.id.taskInput);
```

```
addButton = findViewById(R.id.addButton);
recyclerView = findViewById(R.id.recyclerView);

databaseHelper = new DatabaseHelper(this);
recyclerView.setLayoutManager(new LinearLayoutManager(this));

loadTasks();

addButton.setOnClickListener(v -> {
    String taskName = taskInput.getText().toString().trim();

    if (taskName.isEmpty()) {
        taskInput.setError("Enter a task");
        return;
    }

    boolean inserted = databaseHelper.insertTask(taskName);

    if (inserted) {
        taskInput.setText("");
        loadTasks();
        Toast.makeText(this, "Task added", Toast.LENGTH_SHORT).show();
    } else {
        Toast.makeText(this, "Unable to add task",
```

```
                    Toast.LENGTH_SHORT).show();
            }
        });
    }

    private void loadTasks() {
        taskList = databaseHelper.getAllTasks();

        taskAdapter = new TaskAdapter(taskList, task -> {
            databaseHelper.deleteTask(task.getId());
            loadTasks();
            Toast.makeText(this, "Task deleted", Toast.LENGTH_SHORT).show();
        });

        recyclerView.setAdapter(taskAdapter);
    }
}
```

```
// File: Task.java

package com.example.demoapplication;
```

```
public class Task {

    private final int id;
    private final String taskName;

    public Task(int id, String taskName) {
        this.id = id;
        this.taskName = taskName;
    }

    public int getId() {
        return id;
    }

    public String getTaskName() {
        return taskName;
    }
}


// File: DatabaseHelper.java

package com.example.demoapplication;
```

```
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import java.util.ArrayList;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "TodoDatabase.db";
    private static final int DATABASE_VERSION = 1;

    private static final String TABLE_TASK = "Task";
    private static final String COLUMN_ID = "id";
    private static final String COLUMN_NAME = "task_name";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String query = "CREATE TABLE " + TABLE_TASK + " ("
                + COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, "
```

```
            + COLUMN_NAME + " TEXT NOT NULL)";
    db.execSQL(query);
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_TASK);
    onCreate(db);
}

public boolean insertTask(String taskName) {
    SQLiteDatabase db = this.getWritableDatabase();

    ContentValues values = new ContentValues();
    values.put(COLUMN_NAME, taskName);

    long result = db.insert(TABLE_TASK, null, values);
    db.close();

    return result != -1;
}

public ArrayList<Task> getAllTasks() {
    ArrayList<Task> taskList = new ArrayList<>();
```

```
        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery(
                "SELECT * FROM " + TABLE_TASK + " ORDER BY " + COLUMN_ID
+ " DESC",
                null);


        try {
            while (cursor.moveToNext()) {
                int id = cursor.getInt(cursor.getColumnIndexOrThrow(COLUMN_ID));
                String name = cursor.getString(
                        cursor.getColumnIndexOrThrow(COLUMN_NAME));


                taskList.add(new Task(id, name));
            }
        } finally {
            cursor.close();
            db.close();
        }


        return taskList;
    }


    public boolean deleteTask(int id) {
```

```
        SQLiteDatabase db = this.getWritableDatabase();

        int result = db.delete(
                TABLE_TASK,
                COLUMN_ID + "=?",
                new String[]{String.valueOf(id)}
        );

        db.close();
        return result > 0;
    }
}


// File: TaskAdapter.java

package com.example.demoapplication;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.TextView;
import androidx.annotation.NonNull;
```

```
import androidx.recyclerview.widget.RecyclerView;

import java.util.ArrayList;


public class TaskAdapter extends RecyclerView.Adapter<TaskAdapter.TaskViewHolder> {


    public interface OnDeleteClickListener {

        void onDeleteClick(Task task);

    }


    private final ArrayList<Task> taskList;

    private final OnDeleteClickListener listener;


    public TaskAdapter(ArrayList<Task> taskList,

                       OnDeleteClickListener listener) {

        this.taskList = taskList;

        this.listener = listener;

    }


    @NonNull

    @Override

    public TaskViewHolder onCreateViewHolder(@NonNull ViewGroup parent,

                                             int viewType) {

        View view = LayoutInflater.from(parent.getContext())
```

```
            .inflate(R.layout.task_item, parent, false);

    return new TaskViewHolder(view);
}

@Override
public void onBindViewHolder(@NonNull TaskViewHolder holder, int position) {
    Task task = taskList.get(position);

    holder.taskName.setText(task.getTaskName());
    holder.taskId.setText("Task ID: " + task.getId());

    holder.deleteButton.setOnClickListener(v ->
            listener.onDeleteClick(task));
}

@Override
public int getItemCount() {
    return taskList.size();
}

static class TaskViewHolder extends RecyclerView.ViewHolder {

    TextView taskName, taskId;
```

```
        Button deleteButton;

        TaskViewHolder(@NonNull View itemView) {
            super(itemView);

            taskName = itemView.findViewById(R.id.taskName);
            taskId = itemView.findViewById(R.id.taskId);
            deleteButton = itemView.findViewById(R.id.deleteButton);
        }
    }
}
```

```
<!-- File: res/layout/activity_main.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="ToDo List"
    android:textSize="28sp"
    android:textStyle="bold"
    android:gravity="center"
    android:padding="12dp" />

<EditText
    android:id="@+id/taskInput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter task"
    android:inputType="textCapSentences" />

<Button
    android:id="@+id/addButton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Add Task" />

<androidx.recyclerview.widget.RecyclerView
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
```

```
        android:layout_height="0dp"
        android:layout_weight="1"
        android:paddingTop="8dp" />

</LinearLayout>



<!-- File: res/layout/task_item.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:padding="12dp"
    android:layout_marginBottom="8dp"
    android:background="#EEEEEE">

    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
```

```
    android:orientation="vertical">

    <TextView
        android:id="@+id/taskName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:textStyle="bold"
        android:textColor="#212121" />

    <TextView
        android:id="@+id/taskId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="12sp"
        android:textColor="#666666" />

</LinearLayout>

<Button
    android:id="@+id/deleteButton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Delete" />
```

```
</LinearLayout>



<!-- File: AndroidManifest.xml -->

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="ToDo List"
        android:supportsRtl="true"
        android:theme="@style/Theme.DemoApplication">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
```

```
        </activity>

    </application>
</manifest>
```

7. Using create a demo for Application theme selection by using ToggleButton Dark/Light Mode and keep remember its preference using shared preferences.

```
// File: MainActivity.java

package com.example.demoapplication;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.TextView;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatDelegate;

public class MainActivity extends AppCompatActivity {
```

```
ToggleButton themeToggle;
TextView title;
SharedPreferences preferences;

@Override
protected void onCreate(Bundle savedInstanceState) {

    preferences = getSharedPreferences("ThemePrefs", MODE_PRIVATE);

    boolean isDark = preferences.getBoolean("DarkMode", false);

    AppCompatDelegate.setDefaultNightMode(
            isDark ? AppCompatDelegate.MODE_NIGHT_YES
                    : AppCompatDelegate.MODE_NIGHT_NO
    );

    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    themeToggle = findViewById(R.id.themeToggle);
    title = findViewById(R.id.title);

    themeToggle.setChecked(isDark);
```

```
        updateText(isDark);

        themeToggle.setOnCheckedChangeListener((buttonView, checked) -> {

            preferences.edit()
                    .putBoolean("DarkMode", checked)
                    .apply();

            AppCompatDelegate.setDefaultNightMode(
                    checked ? AppCompatDelegate.MODE_NIGHT_YES
                            : AppCompatDelegate.MODE_NIGHT_NO
            );

            updateText(checked);
        });
    }

    private void updateText(boolean isDark) {
        title.setText(isDark ? "Dark Mode Enabled" : "Light Mode Enabled");
    }
}



<!-- File: res/layout/activity_main.xml -->
```

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Light Mode Enabled"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp" />

    <ToggleButton
        android:id="@+id/themeToggle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOn="Dark Mode"
        android:textOff="Light Mode"
```

```
        android:checked="false" />

</LinearLayout>



<!-- File: res/values/themes.xml -->

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="Theme.DemoApplication"
        parent="Theme.AppCompat.DayNight.NoActionBar">
        <item name="colorPrimary">#6750A4</item>
        <item name="colorPrimaryDark">#453078</item>
        <item name="colorAccent">#6750A4</item>
    </style>
</resources>



<!-- File: AndroidManifest.xml -->

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
```

```
    <application
        android:allowBackup="true"
        android:label="Theme Selection App"
        android:supportsRtl="true"
        android:theme="@style/Theme.DemoApplication">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>
</manifest>



// File: app/build.gradle
```

```
dependencies {
    implementation 'androidx.appcompat:appcompat:1.7.0'
}
```

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MCA_III

Mobile Application Development

Practical Assignment - II

1. Write a program to fetch the current location using the fused Api in Android and print it in a TextView.

```
// File: MainActivity.java

package com.example.demoapplication;

import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Location;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.annotation.NonNull;
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.ContextCompat;

import com.google.android.gms.location.FusedLocationProviderClient;
import com.google.android.gms.location.LocationServices;

public class MainActivity extends AppCompatActivity {

    TextView locationText;
    Button locationButton;
    FusedLocationProviderClient fusedLocationClient;

    ActivityResultLauncher<String[]> locationPermissionLauncher =
            registerForActivityResult(
                    new ActivityResultContracts.RequestMultiplePermissions(),
                    result -> {
                        Boolean fine =
result.get(Manifest.permission.ACCESS_FINE_LOCATION);
                        Boolean coarse =
result.get(Manifest.permission.ACCESS_COARSE_LOCATION);

                        if (Boolean.TRUE.equals(fine)
                                || Boolean.TRUE.equals(coarse)) {
                            getCurrentLocation();
                        } else {
```

```
                    locationText.setText("Location permission denied");
                }
            });

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    locationText = findViewById(R.id.locationText);
    locationButton = findViewById(R.id.locationButton);

    fusedLocationClient =
            LocationServices.getFusedLocationProviderClient(this);

    locationButton.setOnClickListener(v -> checkPermissionAndGetLocation());
}

private void checkPermissionAndGetLocation() {
    boolean fineLocation = ContextCompat.checkSelfPermission(
            this, Manifest.permission.ACCESS_FINE_LOCATION)
            == PackageManager.PERMISSION_GRANTED;

    boolean coarseLocation = ContextCompat.checkSelfPermission(
```

```
            this, Manifest.permission.ACCESS_COARSE_LOCATION)
            == PackageManager.PERMISSION_GRANTED;

    if (fineLocation || coarseLocation) {
        getCurrentLocation();
    } else {
        locationPermissionLauncher.launch(new String[]{
                Manifest.permission.ACCESS_FINE_LOCATION,
                Manifest.permission.ACCESS_COARSE_LOCATION
        });
    }
}

private void getCurrentLocation() {
    if (ContextCompat.checkSelfPermission(
            this, Manifest.permission.ACCESS_FINE_LOCATION)
            != PackageManager.PERMISSION_GRANTED
            && ContextCompat.checkSelfPermission(
            this, Manifest.permission.ACCESS_COARSE_LOCATION)
            != PackageManager.PERMISSION_GRANTED) {
        return;
    }

    locationText.setText("Fetching location...");
```

```
        fusedLocationClient.getLastLocation()
                .addOnSuccessListener(this, location -> {
                    if (location != null) {
                        displayLocation(location);
                    } else {
                        locationText.setText(
                                "Location unavailable. Enable device location and try again.");
                        Toast.makeText(this, "Unable to fetch location",
                                Toast.LENGTH_SHORT).show();
                    }
                })
                .addOnFailureListener(e ->
                        locationText.setText("Error: " + e.getMessage()));
    }

    private void displayLocation(@NonNull Location location) {
        String result = "Current Location\n\n"
                + "Latitude: " + location.getLatitude() + "\n"
                + "Longitude: " + location.getLongitude();

        locationText.setText(result);
    }
}
```

```
<!-- File: res/layout/activity_main.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Current Location Finder"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp" />

    <Button
        android:id="@+id/locationButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
        android:text="Get Current Location" />

    <TextView
        android:id="@+id/locationText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Location will appear here"
        android:textSize="18sp"
        android:gravity="center"
        android:padding="20dp" />

</LinearLayout>



<!-- File: AndroidManifest.xml -->

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission
android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission
android:name="android.permission.ACCESS_COARSE_LOCATION" />
```

```
    <application
        android:allowBackup="true"
        android:label="Location Finder"
        android:supportsRtl="true"
        android:theme="@style/Theme.DemoApplication">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>
</manifest>



// File: app/build.gradle
```

```
dependencies {
    implementation 'com.google.android.gms:play-services-location:21.3.0'
}
```

2. Create an application to play an audio file and use three buttons to play, pause and stop and use a SeekBar to display the progress of the audio file. Make use of MediaPlayer.

```
// File: MainActivity.java

package com.example.demoapplication;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.widget.Button;
import android.widget.SeekBar;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    MediaPlayer mediaPlayer;
    Button playButton, pauseButton, stopButton;
    SeekBar seekBar;
    TextView durationText;
    Handler handler = new Handler(Looper.getMainLooper());

    Runnable updateSeekBar = new Runnable() {
        @Override
        public void run() {
            if (mediaPlayer != null) {
                seekBar.setProgress(mediaPlayer.getCurrentPosition());
                durationText.setText(
                        formatTime(mediaPlayer.getCurrentPosition())
                                + " / " + formatTime(mediaPlayer.getDuration()));

                if (mediaPlayer.isPlaying()) {
                    handler.postDelayed(this, 500);
                }
            }
        }
    };
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    playButton = findViewById(R.id.playButton);
    pauseButton = findViewById(R.id.pauseButton);
    stopButton = findViewById(R.id.stopButton);
    seekBar = findViewById(R.id.seekBar);
    durationText = findViewById(R.id.durationText);

    mediaPlayer = MediaPlayer.create(this, R.raw.audio);

    if (mediaPlayer == null) {
        Toast.makeText(this, "Unable to load audio file",
                Toast.LENGTH_SHORT).show();
        return;
    }

    seekBar.setMax(mediaPlayer.getDuration());
    durationText.setText("00:00 / " + formatTime(mediaPlayer.getDuration()));

    playButton.setOnClickListener(v -> {
        if (mediaPlayer != null && !mediaPlayer.isPlaying()) {
```

```
        mediaPlayer.start();
        handler.removeCallbacks(updateSeekBar);
        handler.post(updateSeekBar);
    }
});

pauseButton.setOnClickListener(v -> {
    if (mediaPlayer != null && mediaPlayer.isPlaying()) {
        mediaPlayer.pause();
        handler.removeCallbacks(updateSeekBar);
    }
});

stopButton.setOnClickListener(v -> {
    if (mediaPlayer != null) {
        mediaPlayer.pause();
        mediaPlayer.seekTo(0);
        seekBar.setProgress(0);
        handler.removeCallbacks(updateSeekBar);
        durationText.setText(
                "00:00 / " + formatTime(mediaPlayer.getDuration()));
    }
});
```

```
        seekBar.setOnSeekBarChangeListener(new
SeekBar.OnSeekBarChangeListener() {

            @Override

            public void onProgressChanged(SeekBar bar, int progress,

                                    boolean fromUser) {

               if (fromUser && mediaPlayer != null) {

                  durationText.setText(formatTime(progress) + " / "

                        + formatTime(mediaPlayer.getDuration()));

               }

            }


            @Override

            public void onStartTrackingTouch(SeekBar bar) {

               handler.removeCallbacks(updateSeekBar);

            }


            @Override

            public void onStopTrackingTouch(SeekBar bar) {

               if (mediaPlayer != null) {

                  mediaPlayer.seekTo(bar.getProgress());


                  if (mediaPlayer.isPlaying()) {

                     handler.post(updateSeekBar);

                  }
```

```
            }
        }
    });

    mediaPlayer.setOnCompletionListener(mp -> {
        handler.removeCallbacks(updateSeekBar);
        seekBar.setProgress(0);
        durationText.setText(
                "00:00 / " + formatTime(mp.getDuration()));
    });
}

private String formatTime(int milliseconds) {
    int minutes = (milliseconds / 1000) / 60;
    int seconds = (milliseconds / 1000) % 60;
    return String.format(java.util.Locale.getDefault(),
            "%02d:%02d", minutes, seconds);
}

@Override
protected void onDestroy() {
    handler.removeCallbacks(updateSeekBar);

    if (mediaPlayer != null) {
```

```
            mediaPlayer.release();
            mediaPlayer = null;
        }

        super.onDestroy();
    }
}
```

```
<!-- File: res/layout/activity_main.xml -->

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Audio Player"
```

```
    android:textSize="28sp"
    android:textStyle="bold"
    android:layout_marginBottom="30dp" />

<TextView
    android:id="@+id/durationText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="00:00 / 00:00"
    android:textSize="18sp"
    android:layout_marginBottom="12dp" />

<SeekBar
    android:id="@+id/seekBar"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:max="100"
    android:layout_marginBottom="24dp" />

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center">
```

```
        <Button
            android:id="@+id/playButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Play" />

        <Button
            android:id="@+id/pauseButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Pause"
            android:layout_marginStart="8dp" />

        <Button
            android:id="@+id/stopButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Stop"
            android:layout_marginStart="8dp" />

    </LinearLayout>

</LinearLayout>
```

```
<!-- File: AndroidManifest.xml -->

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="Audio Player"
        android:supportsRtl="true"
        android:theme="@style/Theme.DemoApplication">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>
```

```
    </application>
</manifest>
```

3. Create an application to play a video file and use three buttons to play, pause and stop and use a SeekBar to display the progress of the video file.

```
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <VideoView
        android:id="@+id/videoView"
        android:layout_width="match_parent"
        android:layout_height="250dp" />
```

```
<SeekBar
    android:id="@+id/seekBar"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:max="100" />

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center"
    android:orientation="horizontal">

    <Button
        android:id="@+id/btnPlay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Play" />

    <Button
        android:id="@+id/btnPause"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Pause" />
```

```
        <Button
            android:id="@+id/btnStop"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Stop" />

    </LinearLayout>

</LinearLayout>
```

```
 // MainActivity.java
package com.example.demoapplication;

import android.net.Uri;
import android.os.Bundle;
import android.os.Handler;
import android.widget.Button;
import android.widget.SeekBar;
import android.widget.VideoView;
```

```
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    VideoView videoView;
    SeekBar seekBar;
    Button btnPlay, btnPause, btnStop;

    Handler handler = new Handler();

    Runnable updateProgress = new Runnable() {
        @Override
        public void run() {
            if (videoView != null && videoView.isPlaying()) {
                int duration = videoView.getDuration();

                if (duration > 0) {
                    seekBar.setProgress(
                        (int) (videoView.getCurrentPosition() * 100L / duration)
                    );
                }
            }
            handler.postDelayed(this, 500);
        }
```

```
};

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    videoView = findViewById(R.id.videoView);
    seekBar = findViewById(R.id.seekBar);
    btnPlay = findViewById(R.id.btnPlay);
    btnPause = findViewById(R.id.btnPause);
    btnStop = findViewById(R.id.btnStop);

    Uri videoUri = Uri.parse(
        "android.resource://" + getPackageName() + "/" + R.raw.sample_video
    );

    videoView.setVideoURI(videoUri);

    btnPlay.setOnClickListener(v -> {
        if (!videoView.isPlaying()) {
            videoView.start();
        }
        handler.post(updateProgress);
```

```
});

btnPause.setOnClickListener(v -> {
    if (videoView.isPlaying()) {
        videoView.pause();
    }
});

btnStop.setOnClickListener(v -> {
    videoView.stopPlayback();
    videoView.setVideoURI(videoUri);
    seekBar.setProgress(0);
});

videoView.setOnPreparedListener(mp -> seekBar.setProgress(0));

seekBar.setOnSeekBarChangeListener(
    new SeekBar.OnSeekBarChangeListener() {
        @Override
        public void onProgressChanged(
            SeekBar bar, int progress, boolean fromUser) {
            if (fromUser && videoView.getDuration() > 0) {
                int position = (int) (
                    videoView.getDuration() * progress / 100.0
```

```
                );
                videoView.seekTo(position);
            }
        }

        @Override
        public void onStartTrackingTouch(SeekBar bar) {
        }

        @Override
        public void onStopTrackingTouch(SeekBar bar) {
        }
    }
    );
}

@Override
protected void onDestroy() {
    handler.removeCallbacks(updateProgress);
    super.onDestroy();
}
}
```

4. Create a gallery application where you can fetch all the images from your device and display them in grid form.

```
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="8dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Image Gallery"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="12dp" />

    <GridView
```

```
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="3"
        android:verticalSpacing="6dp"
        android:horizontalSpacing="6dp"
        android:stretchMode="columnWidth"
        android:gravity="center" />

</LinearLayout>
```

```
/* MainActivity.java */
package com.example.demoapplication;

import android.Manifest;
import android.content.pm.PackageManager;
import android.database.Cursor;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
```

```
import android.provider.MediaStore;
import android.widget.GridView;
import android.widget.ImageView;
import android.widget.BaseAdapter;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    GridView gridView;
    ArrayList<Uri> imageList = new ArrayList<>();
    ImageAdapter adapter;

    private static final int REQUEST_PERMISSION = 100;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```
    gridView = findViewById(R.id.gridView);
    adapter = new ImageAdapter();
    gridView.setAdapter(adapter);

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        checkPermission(Manifest.permission.READ_MEDIA_IMAGES);
    } else {
        checkPermission(Manifest.permission.READ_EXTERNAL_STORAGE);
    }
}

private void checkPermission(String permission) {
    if (ContextCompat.checkSelfPermission(this, permission)
            == PackageManager.PERMISSION_GRANTED) {
        loadImages();
    } else {
        ActivityCompat.requestPermissions(
                this,
                new String[]{permission},
                REQUEST_PERMISSION
        );
    }
}
```

```
@Override
public void onRequestPermissionsResult(
        int requestCode, String[] permissions, int[] grantResults) {
    super.onRequestPermissionsResult(
            requestCode, permissions, grantResults);

    if (requestCode == REQUEST_PERMISSION
            && grantResults.length > 0
            && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
        loadImages();
    } else {
        Toast.makeText(this, "Permission denied",
                Toast.LENGTH_SHORT).show();
    }
}

private void loadImages() {
    imageList.clear();

    Uri uri = MediaStore.Images.Media.EXTERNAL_CONTENT_URI;

    String[] projection = {
            MediaStore.Images.Media._ID
```

```
};

try (Cursor cursor = getContentResolver().query(
        uri,
        projection,
        null,
        null,
        MediaStore.Images.Media.DATE_ADDED + " DESC")) {

    if (cursor != null) {
        int idColumn = cursor.getColumnIndexOrThrow(
                MediaStore.Images.Media._ID);

        while (cursor.moveToNext()) {
            long id = cursor.getLong(idColumn);

            Uri imageUri = Uri.withAppendedPath(
                    MediaStore.Images.Media.EXTERNAL_CONTENT_URI,
                    String.valueOf(id));

            imageList.add(imageUri);
        }
    }
}
```

```
    adapter.notifyDataSetChanged();

    if (imageList.isEmpty()) {
        Toast.makeText(this, "No images found",
                Toast.LENGTH_SHORT).show();
    }
}

class ImageAdapter extends BaseAdapter {

    @Override
    public int getCount() {
        return imageList.size();
    }

    @Override
    public Object getItem(int position) {
        return imageList.get(position);
    }

    @Override
    public long getItemId(int position) {
        return position;
```

```
}

@Override
public android.view.View getView(
        int position,
        android.view.View convertView,
        android.view.ViewGroup parent) {

    ImageView imageView;

    if (convertView == null) {
        imageView = new ImageView(MainActivity.this);

        int size = (int) (110 * getResources()
                .getDisplayMetrics().density);

        imageView.setLayoutParams(
                new GridView.LayoutParams(size, size));

        imageView.setScaleType(ImageView.ScaleType.CENTER_CROP);
        imageView.setPadding(2, 2, 2, 2);
    } else {
        imageView = (ImageView) convertView;
    }
```

```
            imageView.setImageURI(imageList.get(position));


            return imageView;
        }
    }
}
```

```
<!-- AndroidManifest.xml -->
<uses-permission android:name="android.permission.READ_MEDIA_IMAGES" />
<uses-permission
    android:name="android.permission.READ_EXTERNAL_STORAGE"
    android:maxSdkVersion="32" />
```

5. Create an application to fetch all the contacts on your device and display them in list form with their image, first name, last name, and number.

```
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="8dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Device Contacts"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="12dp" />

    <ListView
        android:id="@+id/contactList"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
```

```
</LinearLayout>
```

```
<!-- contact_item.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:padding="10dp">

    <ImageView
        android:id="@+id/contactImage"
        android:layout_width="60dp"
        android:layout_height="60dp"
        android:scaleType="centerCrop"
        android:src="@android:drawable/ic_menu_myplaces" />

    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
```

```
        android:layout_weight="1"
        android:orientation="vertical"
        android:paddingStart="12dp">

        <TextView
            android:id="@+id/contactName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="18sp"
            android:textStyle="bold" />

        <TextView
            android:id="@+id/contactNumber"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:textSize="15sp" />

    </LinearLayout>

</LinearLayout>
```

```
/* MainActivity.java */

package com.example.demoapplication;


import android.Manifest;

import android.content.pm.PackageManager;

import android.database.Cursor;

import android.net.Uri;

import android.os.Bundle;

import android.provider.ContactsContract;

import android.view.LayoutInflater;

import android.view.View;

import android.view.ViewGroup;

import android.widget.BaseAdapter;

import android.widget.ImageView;

import android.widget.ListView;

import android.widget.TextView;

import android.widget.Toast;


import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;

import androidx.core.content.ContextCompat;


import java.util.ArrayList;
```

```
public class MainActivity extends AppCompatActivity {

    ListView contactList;
    ArrayList<Contact> contacts = new ArrayList<>();
    ContactAdapter adapter;

    private static final int REQUEST_CONTACTS = 101;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        contactList = findViewById(R.id.contactList);
        adapter = new ContactAdapter();
        contactList.setAdapter(adapter);

        if (ContextCompat.checkSelfPermission(
                this, Manifest.permission.READ_CONTACTS)
                == PackageManager.PERMISSION_GRANTED) {
            loadContacts();
        } else {
            ActivityCompat.requestPermissions(
                    this,
```

```
            new String[]{Manifest.permission.READ_CONTACTS},
            REQUEST_CONTACTS
        );
    }
}

@Override
public void onRequestPermissionsResult(
        int requestCode, String[] permissions, int[] grantResults) {
    super.onRequestPermissionsResult(
            requestCode, permissions, grantResults);

    if (requestCode == REQUEST_CONTACTS
            && grantResults.length > 0
            && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
        loadContacts();
    } else {
        Toast.makeText(this, "Contacts permission denied",
                Toast.LENGTH_SHORT).show();
    }
}

private void loadContacts() {
    contacts.clear();
```

```
String[] projection = {
        ContactsContract.Contacts._ID,
        ContactsContract.Contacts.DISPLAY_NAME,
        ContactsContract.Contacts.PHOTO_URI,
        ContactsContract.Contacts.HAS_PHONE_NUMBER
};

try (Cursor cursor = getContentResolver().query(
        ContactsContract.Contacts.CONTENT_URI,
        projection,
        null,
        null,
        ContactsContract.Contacts.DISPLAY_NAME + " ASC")) {

    if (cursor != null) {
        while (cursor.moveToNext()) {
            String id = cursor.getString(
                    cursor.getColumnIndexOrThrow(
                            ContactsContract.Contacts._ID));

            String displayName = cursor.getString(
                    cursor.getColumnIndexOrThrow(
                            ContactsContract.Contacts.DISPLAY_NAME));
```

```
String photo = cursor.getString(
        cursor.getColumnIndexOrThrow(
                ContactsContract.Contacts.PHOTO_URI));

int hasPhone = cursor.getInt(
        cursor.getColumnIndexOrThrow(
                ContactsContract.Contacts.HAS_PHONE_NUMBER));

String number = "";

if (hasPhone > 0) {
   try (Cursor phoneCursor = getContentResolver().query(
           ContactsContract.CommonDataKinds.Phone.CONTENT_URI,
           new String[]{
              ContactsContract.CommonDataKinds.Phone.NUMBER
           },
           ContactsContract.CommonDataKinds.Phone.CONTACT_ID
                 + " = ?",
           new String[]{id},
           null)) {

       if (phoneCursor != null
              && phoneCursor.moveToFirst()) {
```

```
                number = phoneCursor.getString(0);
            }
        }
    }

    String firstName = displayName;
    String lastName = "";

    if (displayName != null) {
        String[] parts = displayName.trim().split("\\s+", 2);
        firstName = parts[0];

        if (parts.length > 1) {
            lastName = parts[1];
        }
    }

    contacts.add(new Contact(
            firstName,
            lastName,
            number,
            photo
    ));
}
```

```
        }
    }

    adapter.notifyDataSetChanged();

    if (contacts.isEmpty()) {
        Toast.makeText(this, "No contacts found",
                Toast.LENGTH_SHORT).show();
    }
}

class Contact {
    String firstName, lastName, number, photo;

    Contact(String firstName, String lastName,
            String number, String photo) {
        this.firstName = firstName;
        this.lastName = lastName;
        this.number = number;
        this.photo = photo;
    }
}

class ContactAdapter extends BaseAdapter {
```

```
@Override
public int getCount() {
   return contacts.size();
}

@Override
public Object getItem(int position) {
   return contacts.get(position);
}

@Override
public long getItemId(int position) {
   return position;
}

@Override
public View getView(
      int position, View convertView, ViewGroup parent) {

   if (convertView == null) {
      convertView = LayoutInflater.from(MainActivity.this)
            .inflate(R.layout.contact_item, parent, false);
   }
```

```
        ImageView image = convertView.findViewById(R.id.contactImage);
        TextView name = convertView.findViewById(R.id.contactName);
        TextView number = convertView.findViewById(R.id.contactNumber);

        Contact contact = contacts.get(position);

        name.setText(contact.firstName + " " + contact.lastName);
        number.setText(contact.number.isEmpty()
              ? "No number" : contact.number);

        if (contact.photo != null) {
           image.setImageURI(Uri.parse(contact.photo));
        } else {
           image.setImageResource(
                 android.R.drawable.ic_menu_myplaces);
        }

        return convertView;
     }
  }
}
```

```
<!-- AndroidManifest.xml -->
<uses-permission android:name="android.permission.READ_CONTACTS" />
```

6. Create an application with an ImageView and buttons for Rotate, Zoom, and Fade animations. Apply the respective animation to the image when each button is clicked.

```
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <ImageView
        android:id="@+id/imageView"
```

```
    android:layout_width="200dp"
    android:layout_height="200dp"
    android:src="@drawable/sample_image"
    android:scaleType="fitCenter"
    android:contentDescription="Animation Image" />

<Button
    android:id="@+id/btnRotate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Rotate" />

<Button
    android:id="@+id/btnZoom"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Zoom" />

<Button
    android:id="@+id/btnFade"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Fade" />
```

```
</LinearLayout>
```

```
/* MainActivity.java */
package com.example.demoapplication;

import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.widget.Button;
import android.widget.ImageView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ImageView imageView;
    Button btnRotate, btnZoom, btnFade;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

imageView = findViewById(R.id.imageView);
btnRotate = findViewById(R.id.btnRotate);
btnZoom = findViewById(R.id.btnZoom);
btnFade = findViewById(R.id.btnFade);

btnRotate.setOnClickListener(v -> {
   Animation animation = AnimationUtils.loadAnimation(
         MainActivity.this, R.anim.rotate);
   imageView.startAnimation(animation);
});

btnZoom.setOnClickListener(v -> {
   Animation animation = AnimationUtils.loadAnimation(
         MainActivity.this, R.anim.zoom);
   imageView.startAnimation(animation);
});

btnFade.setOnClickListener(v -> {
   Animation animation = AnimationUtils.loadAnimation(
         MainActivity.this, R.anim.fade);
   imageView.startAnimation(animation);
```

```
        });
    }
}
```

```
<!-- res/anim/rotate.xml -->
<?xml version="1.0" encoding="utf-8"?>
<rotate xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromDegrees="0"
    android:toDegrees="360"
    android:pivotX="50%"
    android:pivotY="50%"
    android:duration="1000"
    android:repeatCount="0" />
```

```
<!-- res/anim/zoom.xml -->
<?xml version="1.0" encoding="utf-8"?>
<scale xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromXScale="1.0"
```

```
    android:toXScale="1.5"
    android:fromYScale="1.0"
    android:toYScale="1.5"
    android:pivotX="50%"
    android:pivotY="50%"
    android:duration="1000"
    android:repeatCount="1"
    android:repeatMode="reverse" />
```

```
<!-- res/anim/fade.xml -->
<?xml version="1.0" encoding="utf-8"?>
<alpha xmlns:android="http://schemas.android.com/apk/res/android"
    android:fromAlpha="1.0"
    android:toAlpha="0.1"
    android:duration="1000"
    android:repeatCount="1"
    android:repeatMode="reverse" />
```

7. Create an quiz application and play sound on correct answer

and different sound on wrong answer

```
<!-- activity_main.xml -->
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Quiz Application"
        android:textSize="26sp"
        android:textStyle="bold"
        android:gravity="center"
        android:padding="16dp" />

    <TextView
        android:id="@+id/txtQuestion"
        android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:text="1. Which language is used for Android development?"
    android:textSize="20sp"
    android:padding="12dp" />

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical">

    <RadioButton
        android:id="@+id/option1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Java" />

    <RadioButton
        android:id="@+id/option2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="HTML" />

    <RadioButton
```

```
        android:id="@+id/option3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="CSS" />

    <RadioButton
        android:id="@+id/option4"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SQL" />

</RadioGroup>

<Button
    android:id="@+id/btnSubmit"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Submit Answer" />

<TextView
    android:id="@+id/txtResult"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:gravity="center"
```

```
        android:textSize="18sp"
        android:padding="12dp" />

</LinearLayout>
```

```
/* MainActivity.java */
package com.example.demoapplication;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.widget.Button;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView txtQuestion, txtResult;
```

```
RadioGroup radioGroup;
RadioButton option1, option2, option3, option4;
Button btnSubmit;

MediaPlayer correctSound, wrongSound;
int score = 0;
boolean answered = false;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    txtQuestion = findViewById(R.id.txtQuestion);
    txtResult = findViewById(R.id.txtResult);
    radioGroup = findViewById(R.id.radioGroup);
    option1 = findViewById(R.id.option1);
    option2 = findViewById(R.id.option2);
    option3 = findViewById(R.id.option3);
    option4 = findViewById(R.id.option4);
    btnSubmit = findViewById(R.id.btnSubmit);

    correctSound = MediaPlayer.create(this, R.raw.correct);
    wrongSound = MediaPlayer.create(this, R.raw.wrong);
```

```
btnSubmit.setOnClickListener(v -> {
    if (answered) {
        return;
    }

    int selectedId = radioGroup.getCheckedRadioButtonId();

    if (selectedId == -1) {
        Toast.makeText(this, "Please select an answer",
                Toast.LENGTH_SHORT).show();
        return;
    }

    answered = true;

    if (selectedId == R.id.option1) {
        score++;
        txtResult.setText("Correct Answer!");
        playSound(correctSound);
    } else {
        txtResult.setText("Wrong Answer!");
        playSound(wrongSound);
    }
```

```
        btnSubmit.setText("Finish");
        btnSubmit.setOnClickListener(view ->
                Toast.makeText(this, "Your Score: " + score + "/1",
                        Toast.LENGTH_LONG).show());
    });
}

private void playSound(MediaPlayer player) {
    if (player != null) {
        if (player.isPlaying()) {
            player.pause();
        }
        player.seekTo(0);
        player.start();
    }
}

@Override
protected void onDestroy() {
    if (correctSound != null) {
        correctSound.release();
        correctSound = null;
    }
```

```
        if (wrongSound != null) {
            wrongSound.release();
            wrongSound = null;
        }

        super.onDestroy();
    }
}
```